QNX Everywhere

PI-SQUARE TECH EXCEEDING EXPECTATIONS

QNX eHACK 2026

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National Level Hackathon

zenith

IOT Edge Guardian

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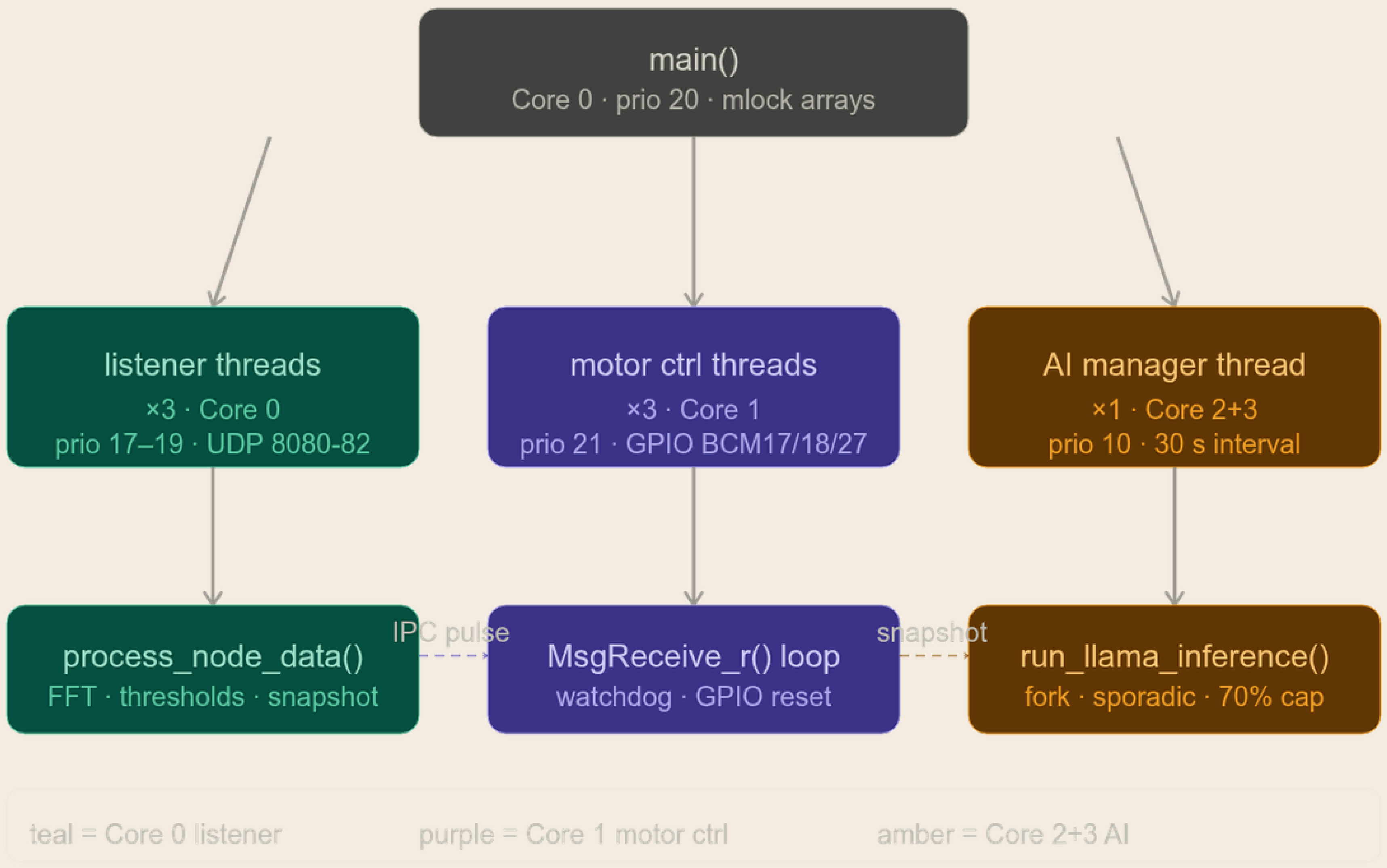
Adithya Y

Section 01.

Design and Execution



Title & Team Identity

Project Name: Iot Edge guardian

Theme Selected: Industrial IoT – Smart Factory "Edge Guardian"

Team Name & City: Team Zenith Mysuru

Two/Three-Line Value Proposition: By leveraging the deterministic microkernel of QNX SDP 8.0, we transform the commercial Raspberry Pi 5 into a hard real-time industrial controller. Our strict CPU core isolation guarantees zero-jitter sensor tracking and microsecond-perfect motor fail-safes, completely eliminating latency spikes even while running heavy predictive AI models on the exact same silicon.

Problem Synthesis & Scenario

The Scenario Unmonitored high-speed industrial motors can experience catastrophic mechanical failure (e.g., shattered bearings) within milliseconds of a fault.
Criticality Emergency shutdown (E-Stop) commands must be executed the exact millisecond a critical vibration threshold (>4.5 RMS) is breached. If the OS scheduler pauses the sensor thread for even 50ms to process a heavy AI workload, the delayed GPIO reset gives the motor enough time to completely tear itself apart.
Success MetricZero-Jitter Hardware Intercept: Guaranteed sub-millisecond latency from UDP sensor trigger to physical GPIO reset, maintaining 100% RTOS thread isolation from the AI diagnostic workload.

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QNX Architectural Approach

Microkernel Strategy (Memory & Comm)

- **Zero Page-Faults: Critical sensor data and state are pinned in physical RAM (mlock()) to prevent latency spikes under heavy AI memory pressure.**
- **Asynchronous IPC: High-frequency control loops use non-blocking messaging (MsgSendPulse_r) to guarantee the RTOS path never stalls waiting for the LLM**

Resource Isolation (CPU Pinning)

- **Hardware Partitioning: Cores 0 & 1 are dedicated exclusively to UDP ingestion and GPIO actuation (ThreadCtl).**
- **Strict AI Sandboxing: The AI inference engine and all its sub-threads are permanently trapped on Cores 2 & 3 using Runmask Inheritance.**

Priority & Scheduling (Execution)

- **Hard Real-Time (SCHED_FIFO): Strict hierarchy prioritizing physical safety: Watchdog (Prio 22) > Motor Control (21) > Sensors (17-19).**
- **Sporadic AI Throttling (SCHED_SPORADIC): The LLM is capped at a 7ms budget per 10ms window (70%), breaking memory bus saturation and preventing cache starvation.**

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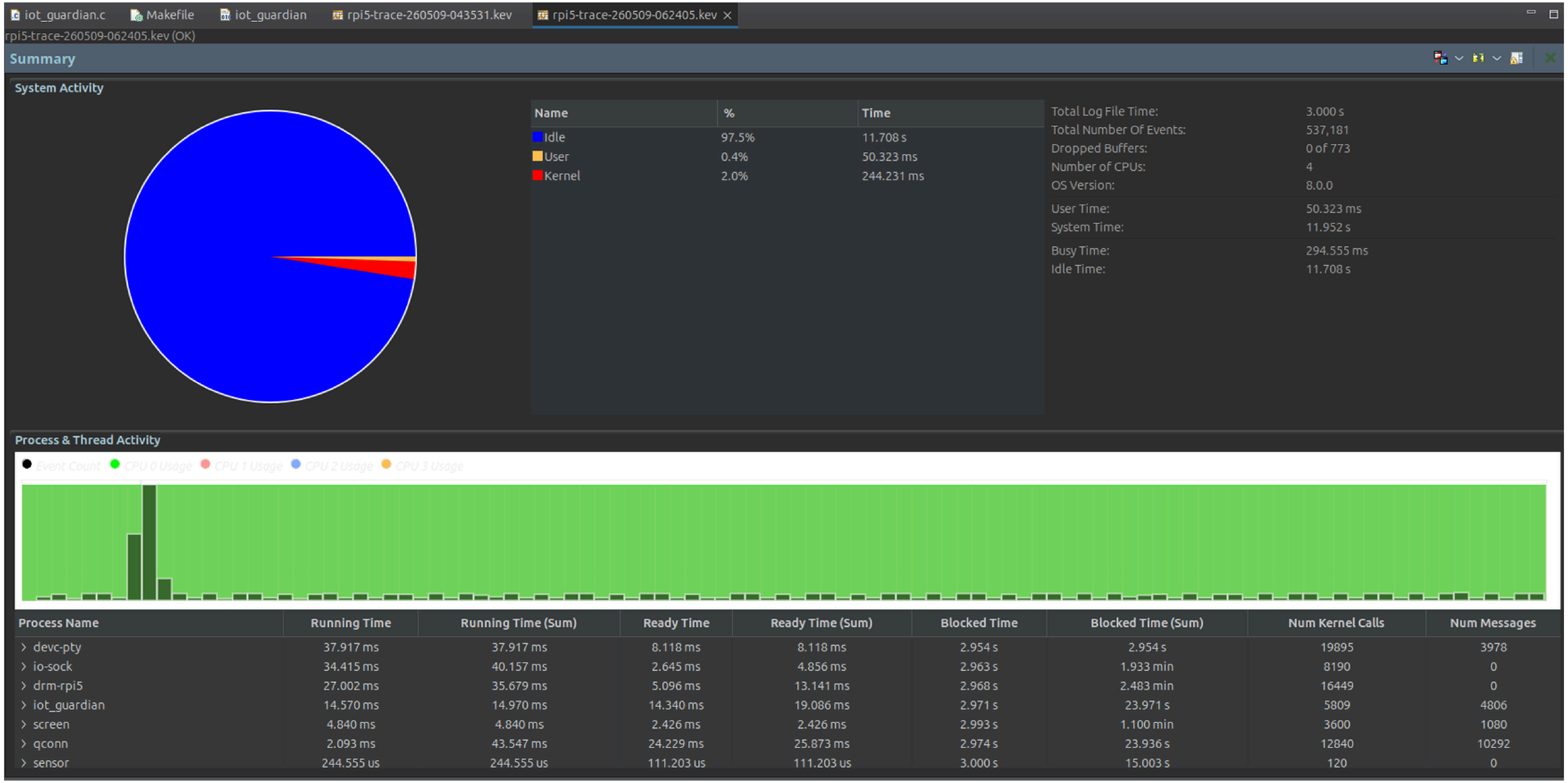
Performance Proof (The Profiler Slide)



Name	%	Time
Idle	97.5%	11.708 s
User	0.4%	50.323 ms
Kernel	2.0%	244.231 ms

Total Log File Time:	3.000 s
Total Number Of Events:	537,181
Dropped Buffers:	0 of 773
Number of CPUs:	4
OS Version:	8.0.0
User Time:	50.323 ms
System Time:	11.952 s
Busy Time:	294.555 ms
Idle Time:	11.708 s

Process Name	Running Time	Running Time (Sum)	Ready Time	Ready Time (Sum)	Blocked Time	Blocked Time (Sum)	Num Kernel Calls	Num Messages
devc-pty	37.917 ms	37.917 ms	8.118 ms	8.118 ms	2.954 s	2.954 s	19895	3978
io-sock	34.415 ms	40.157 ms	2.645 ms	4.856 ms	2.963 s	1.933 min	8190	0
drm-rpi5	27.002 ms	35.679 ms	5.096 ms	13.141 ms	2.968 s	2.483 min	16449	0
iot_guardian	14.570 ms	14.970 ms	14.340 ms	19.086 ms	2.971 s	23.971 s	5809	4806
screen	4.840 ms	4.840 ms	2.426 ms	2.426 ms	2.993 s	1.100 min	3600	1080
qconn	2.093 ms	43.547 ms	24.229 ms	25.873 ms	2.974 s	23.936 s	12840	10292
sensor	244.555 us	244.555 us	111.203 us	111.203 us	3.000 s	15.003 s	120	0

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Future Scalability & Conclusion

Innovation Arbitrage This architecture is hardware-agnostic. By porting this exact QNX microkernel configuration from the RPi 5 to automotive-grade SoCs (like NXP i.MX8 or NVIDIA Jetson), we can instantly scale this solution to meet ISO 26262 / SIL-3 safety standards for commercial EV drivetrains and heavy manufacturing floors.

Learning Outcome Our biggest hurdle was preventing the 1.1B parameter LLM from saturating the memory bus and starving the critical sensor threads. We overcame this by mastering QNX’s SCHED_SPORADIC policy, successfully sandboxing the AI into strict 7ms execution budgets to guarantee zero-jitter RTOS safety.

Closing Statement We are proving that industrial systems no longer have to choose between advanced AI and hard real-time safety. The future of automation is predictive intelligence operating directly at the edge—where the danger actually happens—without missing a single microsecond.